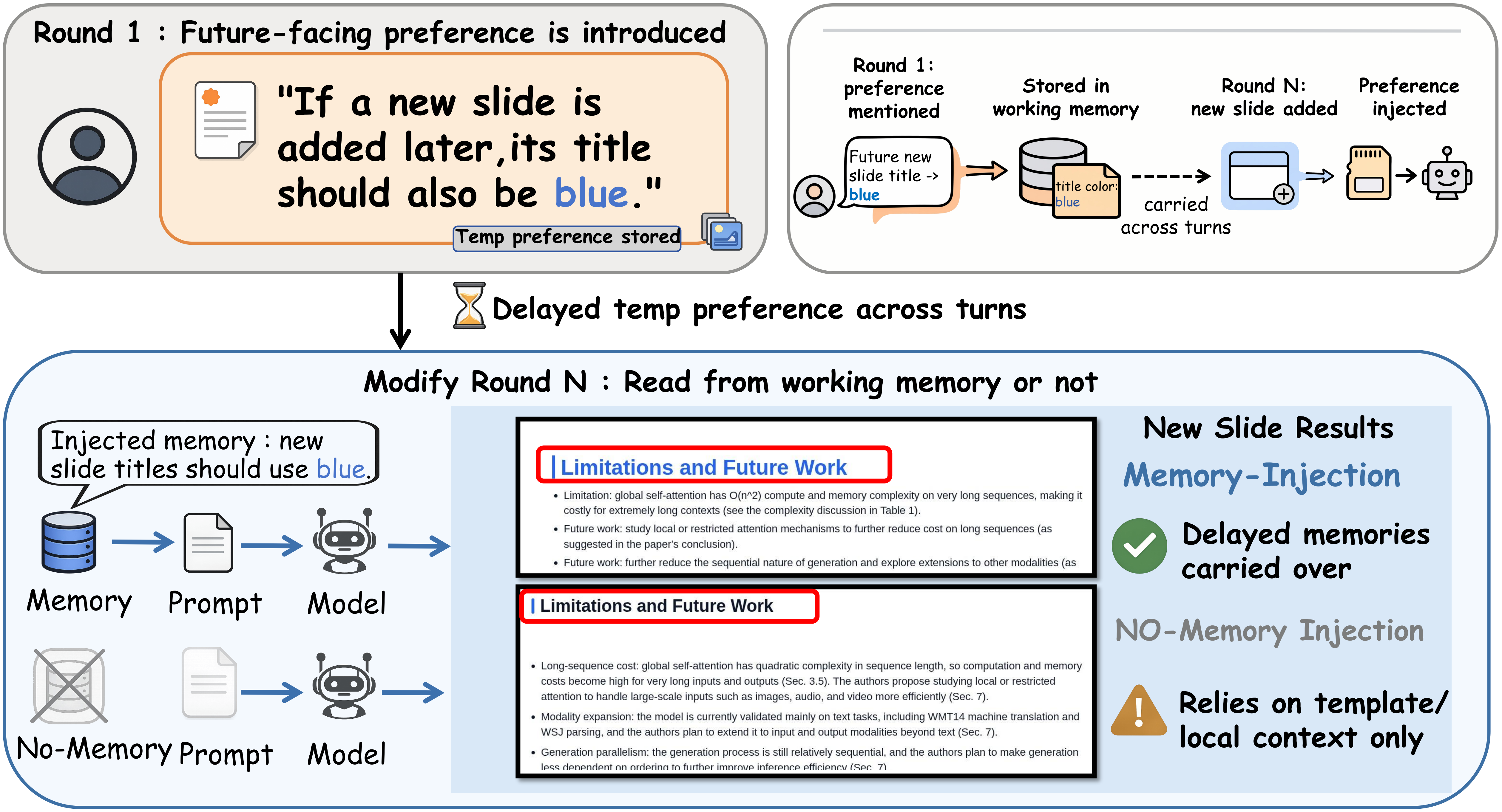

Round 1 : Future-facing preference is introduced
"If a new slide is added later, its title should also be blue."
Temp preference stored
Round 1: preference mentioned
Stored in working memory
Round N: new slide added
Preference injected
Future new slide title -> blue
title color: blue
carried across turns
Delayed temp preference across turns
Modify Round N : Read from working memory or not
Injected memory : new slide titles should use blue.
Memory
Prompt
Model
No-Memory
Prompt
Model
Limitations and Future Work
Limitation: global self-attention has O(n^2) compute and memory complexity on very long sequences, making it costly for extremely long contexts (see the complexity discussion in Table 1).
Future work: study local or restricted attention mechanisms to further reduce cost on long sequences (as suggested in the paper's conclusion).
Future work: further reduce the sequential nature of generation and explore extensions to other modalities (as
Limitations and Future Work
Long-sequence cost: global self-attention has quadratic complexity in sequence length, so computation and memory costs become high for very long inputs and outputs (Sec. 3.5). The authors propose studying local or restricted attention to handle large-scale inputs such as images, audio, and video more efficiently (Sec. 7).
Modality expansion: the model is currently validated mainly on text tasks, including WMT14 machine translation and WSJ parsing, and the authors plan to extend it to input and output modalities beyond text (Sec. 7).
Generation parallelism: the generation process is still relatively sequential, and the authors plan to make generation less dependent on ordering to further improve inference efficiency (Sec. 7).
New Slide Results
Memory-Injection
Delayed memories carried over
NO-Memory Injection
Relies on template/ local context only